Email Spam Detector with Self-Learning Feedback

Final Project Submission Documentation

1. Project Overview

This project is an end-to-end Machine Learning web application designed to classify email text as either *Spam* or *Legitimate (Ham)*. It was built to demonstrate a complete ML lifecycle, from data ingestion and model training to web deployment and continuous self-learning.

2. Architecture & Technologies

- **Backend Framework:** Flask (Python)
- **Machine Learning Library:** Scikit-Learn
- **Algorithm:** Random Forest Classifier
- **Feature Extraction:** TF-IDF (Term Frequency-Inverse Document Frequency)
- **Deployment & CI/CD:** Docker, GitHub Actions
- **Testing:** Pytest (100% Pass Rate across 58 unit tests)
- **Frontend Design:** Premium Dark-Mode UI with Glassmorphism and CSS Micro-animations.

3. Dataset & Model Performance

The model is trained on a robust, real-world dataset comprising over **5,000 email samples**.

- **Total Samples:** 5,764
- **Accuracy:** 97.31%
- **Precision:** 93.87%
- **Recall:** 87.93%
- **F1-Score:** 0.9080

4. Key Engineering Features

1. **Explainable AI (XAI):** The application doesn't just predict "Spam"; it exposes the underlying `feature_importances_` array to highlight the exact words (features) that contributed most heavily to the model's decision.
2. **Self-Learning Loop (MLOps):** Users can provide feedback on predictions. If the model misclassifies an email, the user's correction is saved. A dedicated `/retrain` endpoint dynamically re-ingests this feedback, combining it with the base dataset to retrain the model and vectorizer on the fly.
3. **Robust Error Handling:** Comprehensive input validation and structured logging ensure the application handles malformed data gracefully without crashing.
4. **Containerization:** The entire application, including its dependencies, is containerized using Docker, ensuring perfectly reproducible environments across any system.

5. Conclusion

This project successfully demonstrates advanced software engineering, applied machine learning, and modern UI/UX capabilities, making it a highly production-ready portfolio asset.